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Endoscope and endoscope manufacturing method

Abstract

An endoscope includes: a first substrate includes: a plurality of first pads; a plurality of first cables each including an one end, the one end connected to the plurality of first pads; a plurality of second pads connected to a connection body; and a wiring adjustment circuit disposed between the plurality of first pads and the plurality of second pads. The wiring adjustment circuit is configured to electrically connect each of the plurality of first pads to each of the plurality of second pads corresponding to each of the plurality of first cables.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is based on and claims priority under 35 U.S.C. § 119 to U.S. Provisional Application No. 63/491,968, filed Mar. 24, 2023, the entire contents of which are incorporated herein by reference.

BACKGROUND

(1) The present disclosure relates to an endoscope and an endoscope manufacturing method.
(2) Conventionally, in a method of manufacturing an imaging device mounted on an endoscope shown in Patent Document 1, arranging cable core wires and setting jigs during a cable connection process is complicated. As a result, since the required time becomes longer and the cost increases, there is room for improvement in this respect. [Patent Document 1] Japanese Unexamined Patent Application, First Publication No. 2020-110483

SUMMARY

(3) An endoscope according to a first aspect of the present disclosure is an endoscope including: a first substrate including: a plurality of first pads; a plurality of second pads configured to be connected to a connection body; and a wiring adjustment circuit disposed between the plurality of first pads and the plurality of second pads, and a plurality of first cables, each of the plurality of first cables including a first end, the first end connected to the plurality of first pads. The wiring

adjustment circuit is configured to electrically connect each of the plurality of first pads to each of the plurality of second pads corresponding to each of the plurality of first cables.

(4) An endoscope manufacturing method according to a second aspect of the present disclosure is a method of manufacturing the endoscope including: providing a plurality of first pads and a plurality of second pads on a first substrate; connecting a first end of each of a plurality of first cables to each of the plurality of first pads; and adjusting a wiring between the plurality of first pads and the plurality of second pads according to a wiring pattern of the plurality of first cables on the plurality of first pads by a wiring adjustment circuit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an endoscope system according to a first embodiment.
- (2) FIG. 2 is a plan view showing an outline of a connection portion of a distal end portion and a proximal end portion of an endoscope according to the first embodiment.
- (3) FIG. 3 is a plan view showing an outline of a connection portion of an endoscope according to a first modified example.
- (4) FIG. 4 is a plan view showing a configuration of an imaging device at a proximal end portion.
- (5) FIG. 5 is a plan view showing an outline of a connection portion of an endoscope according to a second embodiment.
- (6) FIG. 6 is a plan view showing an outline of a connection portion of an endoscope according to a third embodiment.
- (7) FIG. 7 is a plan view showing an outline of a connection portion of an endoscope according to a fourth embodiment.
- (8) FIG. 8 is a plan view showing an outline of a connection portion of an endoscope according to a second modified example.

DETAILED DESCRIPTION

First Embodiment

- (9) An endoscope and an endoscope manufacturing method according to a first embodiment of the present disclosure will be described with reference to FIGS. 1 to 4.
- (10) FIG. 1 is a perspective view of an endoscope system **100**.
- (11) The endoscope system **100** includes an endoscope **1**, a universal cord **13**, an external processor unit **50**, and a display device **14**. The external processor unit **50** includes a control unit **50A** and a light source device **50B** and are connected to the endoscope **1** via the universal cord **13**.
- (12) The endoscope **1** is a device for observing or treating an affected area within the body. The endoscope **1** includes an insertion section **15** and an operation section **16**. Furthermore, in this embodiment, the endoscope **1** is a flexible endoscope, but the endoscope **1** may be another type of endoscope (for example, a rigid endoscope or an ultrasonic endoscope).
- (13) The insertion section **15** is a long tubular member that is inserted into the body. The insertion section **15** includes a rigid distal end portion **11**, a bending portion **15A** which can be bent in a plurality of different directions, and a flexible tube portion **15B** having flexibility. The distal end portion **11**, the bending portion **15A**, and the flexible tube portion **15B** are connected in order from the distal end side. The flexible tube portion **15B** is connected to the operation section **16**.
- (14) The distal end portion **11** includes a scope **17**. The scope **17** includes a light source, an optical system, and a camera unit (imaging unit **10**). The bending portion **15A** is bent by the operator's operation of the operation section **16**. The flexible tube portion **15B** is a flexible tubular portion.
- (15) The operation section **16** receives an operation for the endoscope **1**. The universal cord **13** is connected to the operation section **16**.
- (16) The external processor unit **50** comprehensively controls the entire endoscope system **100**. The

external processor unit **50** performs image processing on a video signal output from the imaging unit **10**. The light source device **50B** of the external processor unit **50** supplies illumination light emitted by the light source. The light source device **50B** includes, for example, a halogen lamp or an LED. The light source device **50B** supplies the generated illumination light to the light source under the control of the control unit **50A**.

(17) The display device **14** displays images of the affected area captured by the endoscope **1** and various information regarding the endoscope system **100**.

(18) The imaging unit **10** and the control unit **50A** are connected by a first connection member **4** (a plurality of first cables). The first connection member **4** is a general term for a transmission path composed of a cable inserted through the insertion section **15** of the endoscope **1**, the universal cord **13**, and the like.

(19) The imaging unit **10** converts a subject image formed via the optical system to generate an imaging signal. The imaging unit **10** outputs the generated imaging signal to the control unit **50A** via the first connection member **4**. Further, the imaging unit **10** receives electric power from the control unit **50A** via another first connection member **4**.

(20) As shown in FIG. 2, the endoscope **1** according to the first embodiment includes the distal end portion **11** which is inserted into a subject and a proximal end portion **12** which is connected to the external processor unit **50**. As described above, the external processor unit **50** is disposed outside the endoscope **1** and includes a drive circuit or a video processor.

(21) Furthermore, the external processor unit **50** is not limited to one including a drive circuit or a video processor and may be the universal cord **13** attached to the operation section **16** shown in FIG. 1. The universal cord **13** connects the endoscope **1** and the external device. An imaging cable, optical fiber cable, or the like that outputs the imaging signal captured by the imaging unit **10** to the outside is inserted into the universal cord. In this embodiment, the external processor unit **50** is the video processor.

(22) Here, in the endoscope **1** shown in FIG. 2, the directions used in the description in this embodiment are set such that the right to left direction of the paper is the front to rear direction **X1** and the up to down direction of the paper is the width direction **X2**. Further, in the front to rear direction **X1**, the side of the distal end portion **11** (the side of the imaging unit **10** to be described later) is referred to as the front side (distal end side) and the side of the proximal end portion **12** (the side of the external processor unit **50**) is referred to as the rear side.

(23) The endoscope **1** is a known flexible endoscope, for example, a camera that is a medical flexible endoscope. The endoscope **1** includes an elongated insertion section which extends in the front to rear direction **X1** and is inserted into an observation object from its distal end and a connector **3** which is electrically connected to a rear portion of the insertion section. The endoscope **1** can transmit and receive electric power and various signals (video signals, control signals, and the like) to and from the external processor unit **50**. Furthermore, the endoscope **1** is not limited to a medical camera, and may be an industrial camera.

(24) The video processor **50** has a socket portion (not shown) on the front side and the connector **3** provided at the proximal end portion **12** of the endoscope **1** is connected to this socket portion. The external processor unit **50** performs well-known image processing and the like on still images or moving images obtained by photographing the inside of the observation object (subject).

(25) The above-described electric power and various control signals are guided from the connector **3** to the insertion section **15** via the plurality of first connection members **4** inserted into the insertion section **15** (see FIG. 1). The data of the captured image output from the imaging unit **10** provided at the distal end portion **11** is electrically connected and transmitted from the connector **3** to the external processor unit **50** via the first connection member **4**. The external processor unit **50** performs well-known image processing such as color correction and gradation correction on the data of the captured image transmitted from the connector **3**, and outputs the data of the captured image after the image processing on the display device **14** (see FIG. 1). As described above, the

display device **14** is a monitor device having a display device such as a liquid crystal display panel, and displays the data of the captured image of the subject captured by the endoscope **1**.

(26) The imaging unit **10** is provided to capture an image of the observation object, includes an imaging device such as a CCD or CMOS, and can capture an image of an area to be treated.

(27) The endoscope **1** is formed with a small diameter so that the endoscope can be inserted into the subject. Furthermore, examples of the subject in this embodiment include human blood vessels, ureters, pancreatic ducts, bile ducts, bronchioles, and the like.

(28) The distal end portion **11** is provided with the imaging unit **10** and a front circuit board **20** (second substrate) having a plurality of third pads **21** connected to the imaging unit **10**.

(29) One surface of the front circuit board **20** in the thickness direction is formed as a mounting surface **20a**. The mounting surface **20a** is provided with the plurality of first pads **21**. The front circuit board **20** is formed into a rectangular shape when viewed from a direction orthogonal to the mounting surface **20a**.

(30) The plurality of third pads **21** (five in the illustrated example of FIG. 2) are arranged on the front circuit board **20**. The plurality of third pads **21** are arranged on the rear portion side of the front circuit board **20** at predetermined intervals in the width direction **X2**. Each third pad **21** is used for circuit connection such as a power supply, an image signal, GND (ground), CLK (for example, clock and command input from the external processor unit **50**), and the like. The plurality of third pads **21** are set in an area with a size and shape that allow connection to the core wires **41** of any of the first connection members **4**. Furthermore, the number and arrangement of the third pads **21** are not limited thereto.

(31) Each third pad **21** is electrically connected to any one of the plurality of (here, five) first connection members **4** (**4A**, **4B**, **4C**, **4D**, and **4E**) by soldering. In each first connection member **4**, the outer periphery of the core wire **41** is covered with a coating material. The width of each third pad **21** is set to be equal to or larger than the diameter of the first connection member **4** that is the thickest one among the plurality of first connection members **4**. Furthermore, the plurality of first connection members **4** are connected with an extension direction orthogonal to the width direction **X2** of the third pad **21** and a first pad **32** to be described later.

(32) The first connection members **4** (**4A**, **4B**, **4C**, **4D**, **4E**) are classified into different colors. For example, **4A** is red, **4B** is orange, **4C** is blue, **4D** is black, and **4E** is yellow. A rear end **4b** (one end) of the first connection member **4** is introduced into the connector **3**. The core wires **41** of the plurality of first connection members **4** introduced into the connector **3** are soldered to a predetermined circuit terminal (first pad **32** described later) of a rear circuit board **30** provided in the connector **3** of the proximal end portion **12**. Accordingly, the data of the captured image output from the imaging unit **10** is transmitted from the connector **3** to the external processor unit **50** via the first connection member **4**.

(33) The proximal end portion **12** is provided with the rear circuit board **30** (first substrate) including the plurality of first pads **32** which are connected to the plurality of third pads **21** by the plurality of first connection members **4** and a plurality of second pads **33** which connect the plurality of first pads **32** to the external processor unit **50**. The rear circuit board **30** is provided with a wiring alignment mechanism **60** (wiring adjustment circuit) between the plurality of first pads **32** and the plurality of second pads **33**.

(34) One surface of the rear circuit board **30** in the thickness direction is formed as a mounting surface **30a**. The plurality of first pads **32**, the plurality of second pad **33**, and the wiring alignment mechanism **60** are provided on the mounting surface **30a**. The rear circuit board **30** is formed into a rectangular shape when viewed from a direction orthogonal to the mounting surface **30a**.

(35) The plurality of first pads **32** (five in the illustrated example of FIG. 2) are arranged on the rear circuit board **30**. The plurality of first pads **32** are arranged at predetermined intervals in the width direction **X2** on the front side of the rear circuit board **30**. Each first pad **32** is used for circuit connection such as a power supply, an image signal, GND (ground), CLK (for example, clock and

command input from the video processor **50**), and the like. The plurality of first pads **32** are set in an area with a size and shape that allow connection to the core wires **41** of any of the first connection members **4**. Furthermore, the number and arrangement of the first pads **32** are not limited thereto.

(36) Each first pad **32** is electrically connected to any one of the plurality of (here, five) first connection members **4** (**4A**, **4B**, **4C**, **4D**, and **4E**) respectively connected to the third pads **21** by soldering. The width of each first pad **32** is set to be equal to or larger than the diameter of the first connection member **4** that is the thickest one among the plurality of first connection members **4**.

(37) The plurality of second pads **33** (five in the illustrated example of FIG. 2) are arranged on the rear circuit board **30**. Each second pad **33** is used for circuit connection such as a power supply, an image signal, GND (ground), CLK (for example, clock and command input from the video processor **50**), and the like. Furthermore, the number and arrangement of second pads **33** are not limited thereto.

(38) A second connection member **42** (connection body) connected from the proximal end portion **12** to the external processor unit **50** is electrically connected to the second pad **33**.

(39) Furthermore, the second connection member **42** is not limited to a cable having the same structure as the first connection member **4**, but can be a substrate or a connector having a circuit board.

(40) The wiring alignment mechanism **60** includes a first wiring route **R1** and a second wiring route **R2**. The first wiring route **R1** connects a first wiring **V1** connected to one of the plurality of first pads **32** to one of the plurality of second pads **33** corresponding to the first wiring **V1** so that the external processor unit **50** is electrically connected to the first wiring **V1**. The second wiring route **R2** connects a second wiring **V2** connected to one of the plurality of second pads **33** to one of the plurality of first pads **32** corresponding to the second wiring **V2** so that the imaging unit **10** is electrically connected to the second wiring **V2**. The first wiring **V1** is at least one of the plurality of first connection members **4**. The second wiring **V2** is at least one of the second connection members **42**. As described above, the first wiring **V1** includes a signal line, a power source, and a ground.

(41) Furthermore, the wiring alignment mechanism **60** is not limited to one including both the first wiring route **R1** and the second wiring route **R2** as in this embodiment, and may include at least one of the first wiring route **R1** and the second wiring route **R2**. The wiring alignment mechanism **60** can be a circuit.

(42) As the wiring alignment mechanism **60**, an analog mechanism having a circuit installation area (not shown) is adopted as an example. This circuit installation area is provided with a bridge circuit that can be installed to arrange the wiring according to the wiring order of the plurality of randomly wired first connection members **4**. That is, in the wiring alignment mechanism **60**, a bridge circuit of a plurality of wiring patterns using the plurality of first connection members **4** is facilitated in advance.

(43) Specifically, since there are five connection portions (first pad **21**, second pad **32**) on each of the front and rear ends **4a** and **4b** of the first connection member **4**, **120** patterns of analog bridge circuits are prepared in the circuit installation area. Then, when manufacturing the endoscope **1**, the color combinations of the first connection members **4** connected to each of the third pad **21** and the first pad **32** are identified using an imaging device **70** (see FIG. 4) to be described later to acquire the arrangement data of the plurality of first connection members **4**. Then, the wiring of the first connection member **4** connected to the third pad **21** and the first pad **32** in a random order is checked, the bridge circuit corresponding to the wiring pattern is selected, and the bridge circuit is incorporated into the circuit installation area of the wiring adjustment mechanism **60**, thereby obtaining a function of aligning the wirings of the plurality of first connection members **4**.

(44) Further, as another example of the wiring alignment mechanism **60**, a digital mechanism may be adopted that allows a plurality of types of bridge circuits to be switched in the circuit installation

area after connecting the first connection member **4** in order to correspond to various wiring switching. In this case, the wiring alignment mechanism **60** has a configuration in which the wiring can be switched by software using a program such as a relay circuit according to the arrangement data (arrangement pattern of the plurality of first connection members **4** (**4A** to **4E**)) acquired using the above-described imaging device **70**.

(45) Next, a method of manufacturing the endoscope **1** according to this embodiment will be described in detail. Here, a method of connecting the first connection member **4** to the distal end portion **11** and the proximal end portion **12** will be described.

(46) First, as shown in FIG. **2**, the front circuit board **20** in the imaging unit **10** of the distal end portion **11** is placed on a surface plate (not shown) of which a top surface is a horizontal surface, for example, with the mounting surface **20a** facing upward. Then, any one of the first connection members **4A** to **4E** each having the core wire **41** exposed at the front cable end **4a** is randomly arranged to face the third pad **21** and is connected to each of the plurality of third pads **21** of the front circuit board **20** by soldering in a misaligned state.

(47) Next, the rear circuit board **30** in the connector **3** of the proximal end portion **12** is placed on a surface plate (not shown) of which a top surface is a horizontal surface, for example, with the mounting surface **30a** facing upward. Then, any one of the first connection members **4A** to **4E** each having the core wire **41** exposed at the rear cable end **4b** is randomly arranged to face the first pad **32** and is connected to each of the plurality of first pads **32** of the rear circuit board **30** by soldering in a misaligned state.

(48) Furthermore, as in a first modified example shown in FIG. **3**, when a special cable is adopted as a part of the first connection member **4**, both ends of the special cable may be connected to predetermined fixed positions on the third pad **21** and the first pad **32**. For example, when three of the plurality of (five) first connection members **4** (first connection members **4C**, **4D**, and **4E**) are normal diameter cables and the remaining two cables (first connection members **4A** and **4B**) are special large diameter cables, the large diameter cables are connected to the predetermined third and first pads **21** and **32** by soldering and the normal diameter cables are arranged randomly and are connected to the remaining third and first pads **21** and **32** by soldering in a misaligned state.

(49) Further, the plurality of (five in this case) second connection members **42** each having a core wire **41** exposed at one end are connected to the plurality of second pads **33** of the mounting surface **30a** of the rear circuit board **30** placed on the surface plate by soldering. Further, the other ends of the plurality of second connection members **42** are connected to the external processor unit **50**. Accordingly, the plurality of first connection members **4** connected to the imaging unit **10** can be electrically connected to the respective pads **21**, **32**, and **33** and the connection of the first connection member **4** to the external processor unit **50** is completed.

(50) Next, since there are five pads **21** and **32** on each of the front and rear ends **4a** and **4b** of the first connection member **4**, **120** patterns of analog bridge circuits are prepared in the wiring alignment mechanism **60**. Then, the color combinations of the ten first connection members **4A** to **4E** connected to each of the third pad **21** and the first pad **32** are identified using the imaging device **70** as shown in FIG. **4** to acquire the arrangement data of the plurality of first connection members **4**. Then, a bridge circuit that matches the arrangement data (arrangement pattern of the plurality of first connection members **4** (**4A** to **4E**)) acquired by the imaging device **70** is extracted from **120** patterns, and the wiring is switched using a program. Accordingly, in the wiring alignment mechanism **60**, a necessary wiring route is formed by arranging the randomly wired wires according to the first connection members **4**.

(51) In this way, in the method of manufacturing the endoscope **1**, the plurality of (five) first connection members **4** are classified into different colors and the colors of the plurality of first connection members **4** are identified using the imaging device **70** with a camera to acquire the arrangement data of the plurality of first connection members **4**. Then, the arrangement data is input to the wiring alignment mechanism **60**. In the imaging device **70**, the color of the first

connection member **4** connected to each of the pads **21** and **32** is visually recognized. When connecting the first connection member **4** to the pads **21** and **32**, the connection order of the first connection members **4** can be recognized by the camera of the imaging device **70**. This result is input to the wiring alignment mechanism **60**.

(52) As described above, the endoscope **1** of this embodiment includes the distal end portion **11** which is inserted into the subject and the proximal end portion **12** which is connected to the external processor unit **50**. The endoscope **1** includes: the imaging unit **10** which is provided at the distal end portion **11**; the front circuit board **20** which is provided at the distal end portion **11** and is provided with the plurality of third pads **21** connected to the imaging unit **10**; the rear circuit board **30** which is provided at the proximal end portion **12** and is provided with the plurality of first pads **32** connected to the plurality of second pads **21** by the plurality of first connection members **4** and the plurality of second pads **33** connected to the external processor unit **50** by the plurality of second connection members **42**; and the wiring alignment mechanism **60** which is provided between the plurality of first pads **32** and the plurality of second pads **33** in the rear circuit board **30**. The wiring alignment mechanism **60** includes the first wiring route **R1** which connects the first wiring **V1** connected to one of the plurality of first pads **32** to one of the plurality of second pads **33** corresponding to the first wiring **V1** so that the external processor unit **50** is electrically connected to the first wiring **V1** and the second wiring route **R2** which connects the second wiring **V2** connected to one of the plurality of second pads **33** to one of the plurality of first pads **32** corresponding to the second wiring **V2** so that the imaging unit **10** is electrically connected to the second wiring **V2**.

(53) Further, the endoscope manufacturing method of this embodiment includes: providing the plurality of third pads **21** in the front circuit board **20**; providing the plurality of first pads **32** in the rear circuit board **30**; electrically connecting the third pad **21** and the first pad **32** by the plurality of first connection members **4**; electrically connecting the first pad **32** and the second pad **33**; and forming at least one of the first wiring route **R1** and the second wiring route **R2** using the wiring adjustment mechanism **60**.

(54) Accordingly, in this embodiment, in the steps of manufacturing the endoscope **1**, for each of the plurality of third pads **21** of the front circuit board **20** at the distal end portion **11**, any one of the plurality of first connection members **4A** to **4E** each having the core wire **41** exposed at the front cable end **4a** is randomly connected to the third pad **21**. Further, for each of the plurality of first pads **32** of the rear circuit board **30** at the proximal end portion **12**, any one of the plurality of first connection members **4A** to **4E** having the core wire **41** exposed at the rear cable end **4b** is randomly connected to the first pad **32**. Further, the plurality of (here, five) second connection members **42** each having the core wire **41** exposed at one end are connected to each of the plurality of second pads **33** of the mounting surface **30a** of the rear circuit board **30** and the other ends of the plurality of second connection members **42** are connected to the external processor unit **50**.

(55) Accordingly, the plurality of first connection members **4** connected to the imaging unit **10** can be electrically connected to the respective pads **21**, **32**, and **33** and the connection of the first connection member **4** to the external processor unit **50** is completed. Then, the wiring alignment mechanism **60** includes a mechanism for switching the wiring pattern according to the wiring pattern of the first connection members **4** from the proximal end portion side to the distal end portion side and forms at least one of the first wiring route **R1** and the second wiring route **R2**. Accordingly, it is possible to form a necessary wiring route by aligning the wirings according to the first connection members **4** randomly wired.

(56) Here, as a conventional endoscope manufacturing method, semi-automatic cable connection using laser cutting and pulse heating equipment has been proposed with the aim of reducing costs, but full automation has not yet been achieved. In other words, arranging cables and setting jigs is complicated and difficult to automate, and some processes require manual labor. In contrast, in this embodiment, the first connection members **4A** to **4E** can be aligned with the pads **21** and **32** in a

random order. Therefore, the work of arranging the cables becomes unnecessary, the work of connecting the first connection member **4** can be made more efficient, the process can be shortened, and costs can be reduced. Therefore, the wiring alignment mechanism **60** can be used to automate or semi-automate the alignment work.

(57) Further, in this embodiment, the first wiring **V1** is at least one of the plurality of first connection members **4**. The second wiring **V2** is at least one of the plurality of second connection members **42**. Therefore, the first wiring **V1** output from the imaging unit **10** is transmitted to the external processor unit **50** via the first pad **32** and the second pad **33** and the second wiring **V2** output from the external processor unit **50** is transmitted to the imaging unit **10** via the second pad **33**, the first pad **32**, and the third pad **21**.

(58) Further, in this embodiment, the plurality of first connection members **4** connected with the extension direction orthogonal to the width direction **X2** of the third pad **21** and the first pad **32** and the width dimension of each of the third pad **21** and the first pad **32** is equal to or larger than the diameter of the first connection member **4** that is the thickest one among the plurality of first connection members **4**. Therefore, even when connecting the plurality of first connection member **4** having a plurality of different outer diameters, the thickest first connection member **4** can be reliably connected to any of the plurality of pads **21** and **32** among the plurality of pads **21** and **32** without protruding.

(59) Further, in this embodiment, the wiring adjustment mechanism **60** has a circuit installation area in which a bridge circuit that aligns the wiring according to the wiring of the plurality of first connection members **4** can be incorporated. Therefore, in this embodiment, the bridge circuit of the plurality of wiring patterns using the plurality of first connection members **4** is facilitated in advance, the wiring of the randomly connected first connection member **4** is checked, the bridge circuit corresponding to the wiring pattern is selected, and the bridge circuit is incorporated into the circuit installation area of the wiring adjustment mechanism **60**. In this way, the wirings of the plurality of first connection members **4** can be aligned.

(60) Further, in this embodiment, since the wiring adjustment mechanism includes a mechanism for switching the wiring using a program according to the arrangement pattern of the plurality of first connection members **4**, the wiring can be switched by software using a program of a relay circuit or the like according to the acquired arrangement data (arrangement pattern of the plurality of first connection members **4** (**4A** to **4E**)). Therefore, it becomes possible to automate the alignment work more easily.

(61) According to the endoscope **1** and the method of manufacturing the endoscope **1** of this embodiment, it is possible to reduce the work required to align the first connection members **4** when connecting cables and to improve work efficiency, shorten process time, and reduce costs. Further, according to the endoscope **1** and the method of manufacturing the endoscope **1** of this embodiment, cable connections can be automated or semi-automated.

(62) Although the embodiment of the present disclosure has been described above in detail with reference to the drawings, the specific configuration is not limited to this embodiment, and design changes and the like may be made without departing from the spirit of the present disclosure. Further, the components shown in the above-described embodiments and modified examples can be configured by appropriately combining them.

Second Embodiment

(63) As shown in FIG. 5, an endoscope **1A** according to a second embodiment adopts a plurality of (here, five) coaxial cables **40** (**40A**, **40B**, **40C**, **40D**, and **40E**) including a core wire **410** and a shield **420** covering the outer periphery of the core wire **410** as a plurality of first connection members connecting the distal end portion **11** and the proximal end portion **12**. Five coaxial cables **40** are respectively classified into different colors. The shield **420** prevents static electricity from being applied to the core wire **410** from the outside and electrostatic damage to electronic components such as an imaging element of the imaging unit **10**. The shield **420** may be, for

example, a metal foil such as aluminum, a braided shield made of thin conductive wires knitted into a mesh, or a two-layered structure of a metal foil and a braided shield. The outer peripheries of the plurality of coaxial cables **40** are further covered with a tubular cable sheath (not shown) made of soft insulating resin.

(64) The third pad **21** includes a core wire pad **21A** which is connected to the core wire **410** and a shield pad **21B** which is connected to the shield **420**. The pair of core wire pads **21A** and shield pads **21B** are arranged at intervals in the front to rear direction **X1**. A plurality of pairs of core wire pads **21A** and shield pads **21B** are arranged in the width direction **X2**. The first pad **32** includes a core wire pad **32A** which is connected to the core wire **410** and a shield pad **32B** which is connected to the shield **420**. A pair of core wire pads **32A** and shield pads **32B** are arranged at intervals in the front to rear direction **X1**. A plurality of pairs of core wire pads **32A** and shield pads **32B** are arranged in the width direction **X2**.

(65) Even in the endoscope **1A** according to the second embodiment, as in the first connection member **4** (see FIG. 2) of the above-described first embodiment, any one of the coaxial cables **40** each having the core wire **410** and the shield **420** exposed at the front cable end **40a** is randomly arranged and connected to each of the plurality of first pads **21** of the front circuit board **20** in a misaligned state by soldering. The same applies to the connection of a rear cable end **40b** of the coaxial cable **40** to the first pad **32** of the rear circuit board **30**.

(66) In the method of manufacturing the endoscope **1A** according to the second embodiment, since there are five pads **21** (**21A**, **21B**) and **32** (**32A**, **32B**) on each of both front and rear ends **40a** and **40b** of the coaxial cable **40**, **120** patterns of analog bridge circuits are prepared in the wiring alignment mechanism **60**. Then, the color combinations of the ten coaxial cables **40** connected to each of the third pad **21** and the first pad **32** are identified using the imaging device **70** (see FIG. 4) to acquire the arrangement data of the plurality of coaxial cables **40** (**40A** to **40E**). The bridge circuit matching the acquired arrangement data (the arrangement pattern of the plurality of coaxial cables **40**) is extracted from **120** patterns and the wiring is switched using a program. Accordingly, in the wiring alignment mechanism **60**, a necessary wiring route is formed by aligning the wiring according to the randomly wired coaxial cables **40**.

(67) Further, in the second embodiment, at least one (here, one coaxial cable **40A**) of the plurality of (five) coaxial cables **40** on the side of the distal end portion **11** electrically connects the core wire pad **21A** and the shield pad **21B** by a first connection line **22**. Accordingly, in the coaxial cable **40A**, the core wire **410** and the shield **420** can be used for the same function, and can be used as low impedance wiring (for example, power cable).

(68) Further, in the second embodiment, at least two (here, two coaxial cables **40D** and **40E**) of the plurality of (five) coaxial cables **40** on the side of the distal end portion **11** electrically connect the core wire pads **21A** and the shield pads **21B** by a second connection line **23**. Accordingly, the coaxial cables **40D** and **40E** can be used for the same function and can be used as low impedance wiring (for example, ground).

(69) Furthermore, the first connection line **22** and the second connection line **23** can use the core wire **410** and the shield **420** for the same function on the side of the proximal end portion **12** as well as on the side of the distal end portion **11** and can use the plurality of coaxial cables **40** for the same function.

Third Embodiment

(70) As shown in FIG. 6, an endoscope **1B** according to a third embodiment is provided with a wiring adjustment mechanism **61** that allows a wiring (a connection wiring **36** to be described later) to select a conduction destination after connecting the coaxial cable **40** of the second embodiment to each of the pads **21** and **32**.

(71) The wiring adjustment mechanism **61** will be described. In the connector **3**, a plurality of first conduction tip portions **34** corresponding to all of the plurality of coaxial cables **40** (**40A**, **40B**, **40C**, **40D**, and **40E**) are provided for each of the plurality of (five) first pads **32**. Further, an

unwired area **35** is provided in a state before manufacturing between the core wire pads **32A** of the plurality of first pads **32** and the plurality of first conduction tip portions **34** in the front to rear direction **X1**.

(72) The first conduction tip portions **34** are provided at five positions corresponding to the number of coaxial cables **40** for one core pad **32A**. Therefore, the rear circuit board **30** is provided with a total of 25 first conduction tips **34**. The plurality of first conduction tip portions **34** are arranged to protrude forward from the connector **3**.

(73) In the wiring adjustment mechanism **61**, a necessary wiring route is formed by aligning the wirings according to the randomly wired coaxial cable **40** in such a manner that the connection wiring **36** is formed in the unwired area **35** so that the plurality of first pads **32** and one of the plurality of first conduction tip portions **34** are selectively connected according to the wiring of the plurality of coaxial cables **40**. As the connection wiring **36**, for example, solder jet bumping, cream solder, additional jumper wire, and the like can be adopted. When connecting with an additional jumper wire, even if the distance in the width direction **X2** between the first pad **32** selected for connection and the first conduction tip portion **34** is far, the connection work can be performed more efficiently.

(74) In the connection method using the wiring adjustment mechanism **61** according to the third embodiment, the plurality of first pads **32** and one of the plurality of first conduction tip portions **34** are selectively connected according to the wiring of the plurality of coaxial cables **40**. In this way, an arbitrary circuit can be formed by connecting the predetermined core wire pad **32A** and the arbitrarily selected first conduction tip portion **34** in the unwired area **35** by the connection wiring **36**.

Fourth Embodiment

(75) As shown in FIG. 7, an endoscope **1C** of a fourth embodiment is provided with a wiring adjustment mechanism **62** that can select a conduction destination by connecting the coaxial cable **40** of the second embodiment to each of the pads **21** and **32** and cutting an arbitrary circuit using, for example, laser or the like.

(76) In the wiring adjustment mechanism **62**, a plurality of second conduction tip portions **37** corresponding to all of the plurality of coaxial cables **40** (**40A**, **40B**, **40C**, **40D**, and **40E**) are provided for each of the plurality of (five) first pads **32A**. Further, the plurality of first pads **32** are connected to all of the plurality of second conduction tip portions **37**. A plurality of cutting portions **37A** capable of selectively cutting an unnecessary circuit according to the wiring of the plurality of coaxial cables **40** are arranged in one direction (the width direction **X2**).

(77) The second conduction tip portions **37** are provided at five positions corresponding to the number of coaxial cables **40** for one core wire pad **32A**. Therefore, a total of 25 second conduction tips **37** are provided on the rear circuit board **30**. The plurality of second conduction tip portions **37** protrude forward from the connector **3** and are connected to the core wire pad **32A**.

(78) In the connection method using the wiring adjustment mechanism **62** according to the fourth embodiment, a circuit is selectively cut in such a manner that the unnecessary second conduction tip portions **37** are cut by a cutting tool such as a cutter according to the wiring of the plurality of coaxial cables **40** to form the plurality of cutting portions **37A**. That is, an arbitrary circuit can be formed by selectively connecting the plurality of second pads **32A** and one of the plurality of first conduction tip portions **34**.

Second Modified Example

(79) As shown in FIG. 8, a second modified example is a modified example of a method of manufacturing the endoscope **1A** based on the above-described second embodiment. That is, in the second embodiment, the randomly colored coaxial cables **40** are photographed using the imaging device **70** (see FIG. 4), and array data is obtained through image processing. However, in the second modified example, the arrangement data is acquired by measuring impedance characteristics instead of the method using such an imaging device **70**.

(80) In the wiring alignment mechanism **63** (wiring adjustment mechanism), a plurality of test terminals **65** respectively corresponding to the plurality of (five) first pads **32A** are provided in the rear circuit board **30**. The plurality of test terminals **65** are arranged in one direction (the width direction **X2**). In the wiring alignment mechanism **63**, the impedance characteristics of different test terminals **65** are measured using a measuring device **66**.

(81) In the second modified example, when manufacturing the endoscope **1**, the arrangement data of the plurality of coaxial cables **40** is obtained by measuring the impedance characteristics of the plurality of coaxial cables **40** by the electrical test on the base end side of the first pad **32** which is on the side of the connector **3** and the arrangement data is input to the wiring adjustment mechanism **61**. In this way, in the second modified example, since there is no need to classify the colors of the cables, the material cost for the cables can be reduced.

(82) Furthermore, the method of acquiring arrangement data by measuring impedance characteristics according to the second modified example and the method of acquiring arrangement data by performing image processing using the imaging device **70** according to the second embodiment may be used in combination.

(83) Although preferred embodiments and modified examples of the present disclosure have been described above along with examples, the present disclosure is not limited to these embodiments, modified examples, and examples. Additions, omissions, substitutions, and other changes to the configuration are possible without departing from the spirit of the disclosure.

(84) Further, the present disclosure is not limited by the foregoing description, but only by the scope of the claims appended hereto.

(85) The present disclosure can be applied to the endoscope and the endoscope manufacturing method.

Claims

1. An endoscope comprising: a first substrate comprising: a plurality of first pads; a plurality of second pads configured to be connected to a connection body; and a wiring adjustment circuit disposed between the plurality of first pads and the plurality of second pads, a plurality of first cables, each of the plurality of first cables including a first end, the first end connected to the plurality of first pads; wherein the wiring adjustment circuit is configured to electrically connect each of the plurality of first pads to each of the plurality of second pads corresponding to each of the plurality of first cables.
2. The endoscope according to claim 1, further comprising: a distal end portion configured to insert into a subject, the distal end portion comprising: an imager; and a second substrate comprising a plurality of third pads connected to the imager, wherein each of the plurality of first cables includes a second end, each of the second end is connected to each of the plurality of third pads.
3. The endoscope according to claim 2, wherein the plurality of first cables are coaxial cables each including a core wire and a shield, and wherein the plurality of third pads includes a core wire pad connected to the core wire and a shield pad connected to the shield.
4. The endoscope according to claim 3, wherein at least one of the plurality of third pads is provided to electrically connect the core wire pad and the shield pad.
5. The endoscope according to claim 2, wherein an arrangement of the first end of the plurality of first cables is different from an arrangement of the second end of the plurality of first cables.
6. The endoscope according to claim 2, further comprising: the connection body configured to be connected to an external processor, wherein the wiring adjustment circuit is configured to electrically connect from the imager to the external processor.
7. The endoscope according to claim 1, wherein the wiring adjustment circuit includes at least one of a first wiring route and a second wiring route, the first wiring route from one of the plurality of first pads to one of the plurality of second pads corresponding to one of the plurality of first cables,

the second wiring route from one of the plurality of second pads to one of the plurality of first pads to which one of the plurality of first cables corresponding to the one of the plurality of second pads connected.

8. The endoscope according to claim 1, further comprising: the connection body configured to be connected to an external processor.

9. The endoscope according to claim 8, wherein an arrangement of the first end of the plurality of first cables is different from an arrangement of the connection body.

10. The endoscope according to claim 1, wherein a width of the plurality of first pads is orthogonal to a direction in which the plurality of first cables extends, and wherein the width of the plurality of first pads is equal to or larger than an outer diameter of the thickest one among the plurality of first cables.

11. The endoscope according to claim 1, wherein the wiring adjustment circuit includes a circuit installation area in which a bridge circuit configured to be incorporated, the bridge circuit aligns a wiring according to a wiring pattern of the plurality of first cables on the plurality of first pads.

12. The endoscope according to claim 1, wherein the wiring adjustment circuit is configured to switch a wiring using a program according to a wiring pattern of the plurality of first cables on the plurality of first pads.

13. The endoscope according to claim 1, wherein the wiring adjustment circuit includes a plurality of conduction portions corresponding to all of the plurality of first cables for each of the plurality of first pads, wherein the each of the plurality of first pads and one of the plurality of conduction portions are selectively connected according to the plurality of first cables.

14. The endoscope according to claim 13, wherein the each of the plurality of first pads and the one of the plurality of conduction portions is connected by soldering.

15. The endoscope according to claim 13, wherein the each of the plurality of first pads and the one of the plurality of conduction portions is connected by a jumper wire.

16. The endoscope according to claim 1, wherein the wiring adjustment circuit includes a plurality of conduction portions corresponding to all of the plurality of first cables for each of the plurality of first pads, wherein the each of the plurality of first pads are connected to all of the plurality of conduction portions, and wherein a cutting portion disposed in each of the plurality of conduction portions, the cutting portion of an unnecessary circuit among the plurality of conduction portions is configured to selectively cut off according to a wiring pattern of the plurality of first cables connected to the each of the plurality of first pads.

17. A method of manufacturing an endoscope comprising: providing a plurality of first pads and a plurality of second pads on a first substrate; connecting a first end of each of a plurality of first cables to each of the plurality of first pads; and adjusting a wiring between the plurality of first pads and the plurality of second pads according to a wiring pattern of the plurality of first cables on the plurality of first pads by a wiring adjustment circuit.

18. The method of manufacturing the endoscope according to claim 17, further comprising: providing a plurality of third pads on a second substrate; connecting a second end of each of the plurality of first cables to each of the plurality of third pads.

19. The method of manufacturing the endoscope according to claim 17, further comprising: respectively classifying the plurality of first cables into different colors, identifying the colors of the plurality of first cables using an imaging device; acquiring arrangement data of the plurality of first cables by the identifying; and inputting the arrangement data to the wiring adjustment circuit.

20. The method of manufacturing the endoscope according to claim 17, further comprising: measuring impedance characteristics of each of the plurality of first cables at a position between the plurality of first pads and the wiring adjustment circuit; acquiring arrangement data of the plurality of first cables by the measuring; and inputting the arrangement data to the wiring adjustment circuit.
